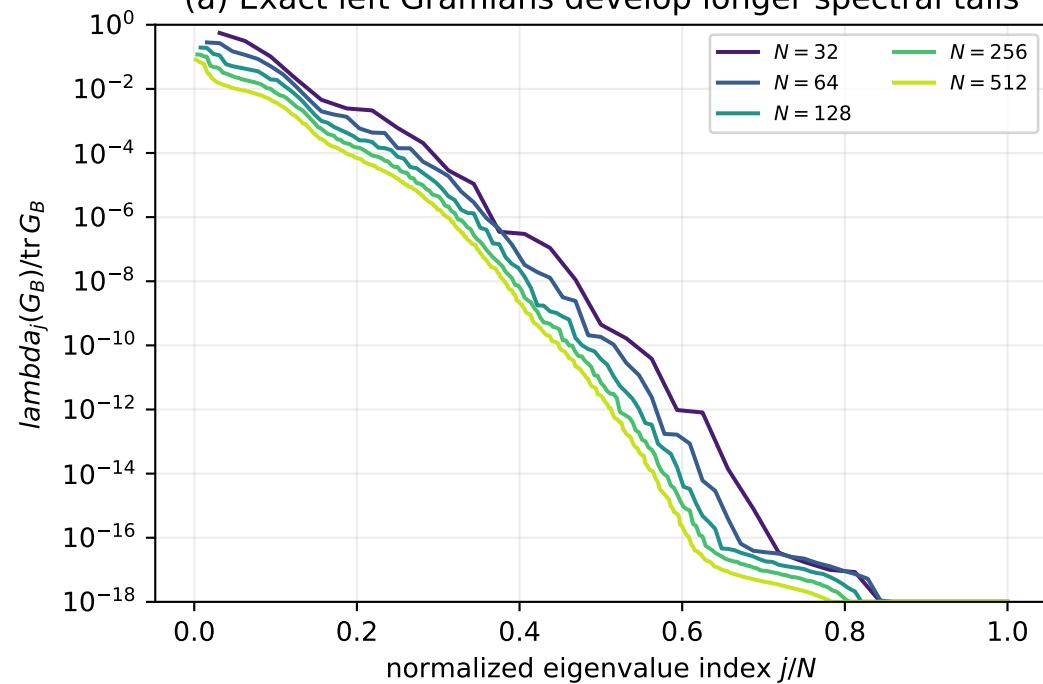
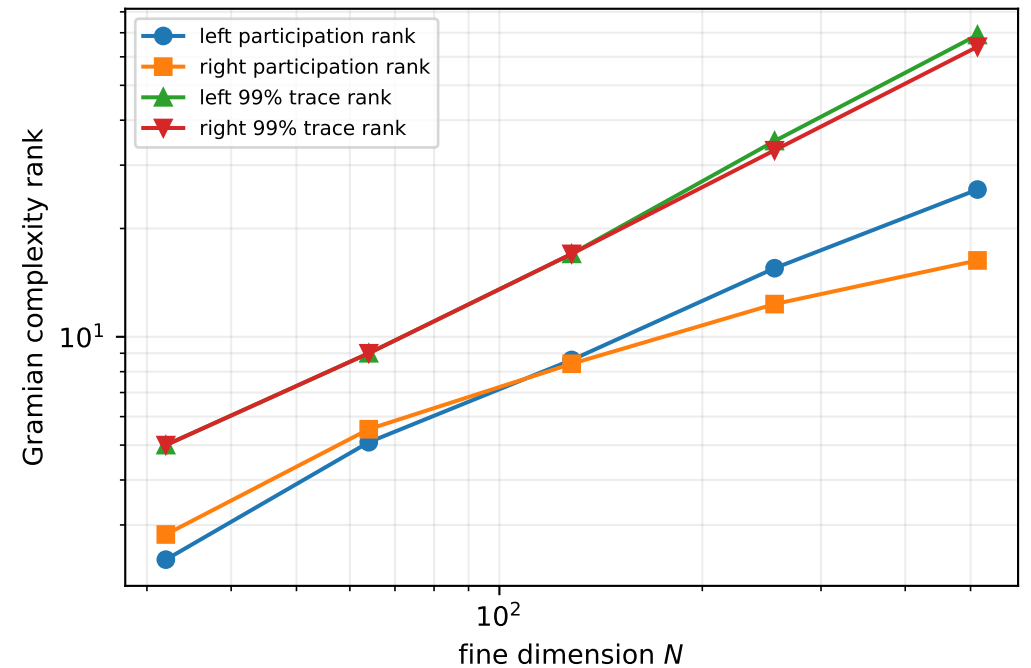


Structured Stein certificates: high-dimensional Gramians and a viable growing-horizon closure

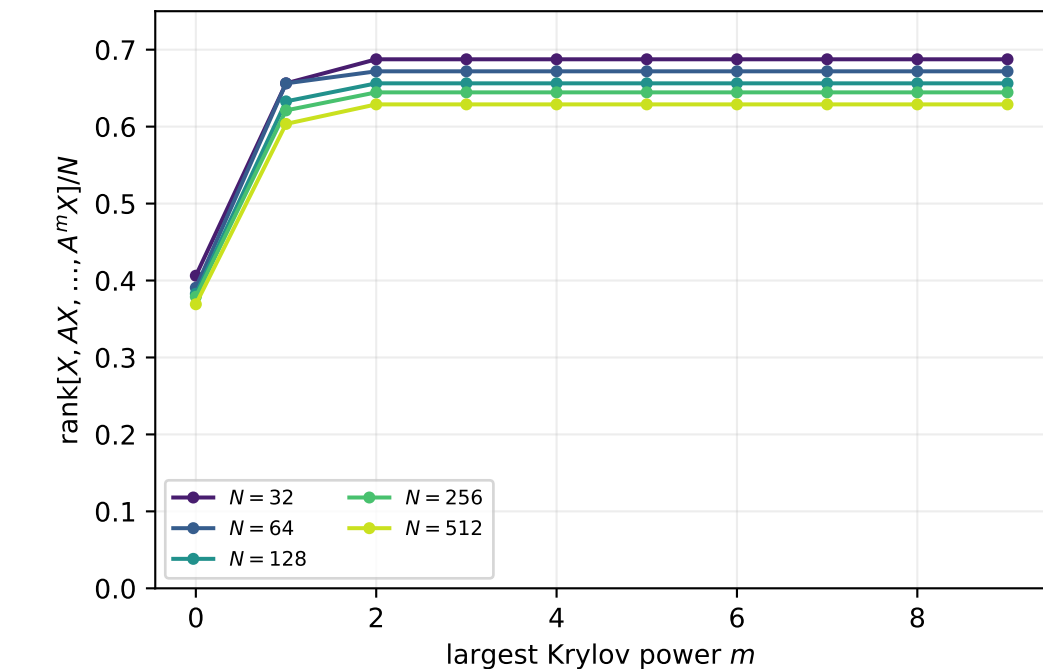
(a) Exact left Gramians develop longer spectral tails



(b) Energy stays moderate while state complexity grows



(c) The directional cyclic span rapidly fills about $0.64N$



(d) A growing horizon restores a sharp positive certificate

